

OPMIG-02: OpenPipeline Migration

Guide: Part 2

Series: OPMIG — OpenPipeline Migration | **Notebook:** 2 of 10 | **Created:** December 2025 | **Last Updated:** 07/20/2026

Architecture & Key Concepts

Table of Contents

1. [Data Flow Architecture](#)
 2. [Processing Stages in Detail](#)
 3. [Pipeline Types](#)
 4. [Processor Types](#)
 5. [Dynatrace Pattern Language \(DPL\)](#)
 6. [Complete OpenPipeline Limits Reference](#)
 7. [Entity Field Availability Timeline](#)
 8. [Key Fields & Metadata](#)
 9. [Exploring Your Pipeline Configuration](#)
 10. [Understanding Processing Order](#)
 11. [Summary: Key Architecture Concepts](#)
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Learning Objectives

By the end of this notebook, you will:

-  Understand the complete OpenPipeline data flow architecture

-  Learn detailed processing stage execution order
-  Master DPL (Dynatrace Pattern Language) fundamentals
-  **Know ALL OpenPipeline limits in comprehensive detail**
-  Understand processor types and capabilities
-  **Learn entity field availability timeline**
-  Explore your pipeline configuration with DQL

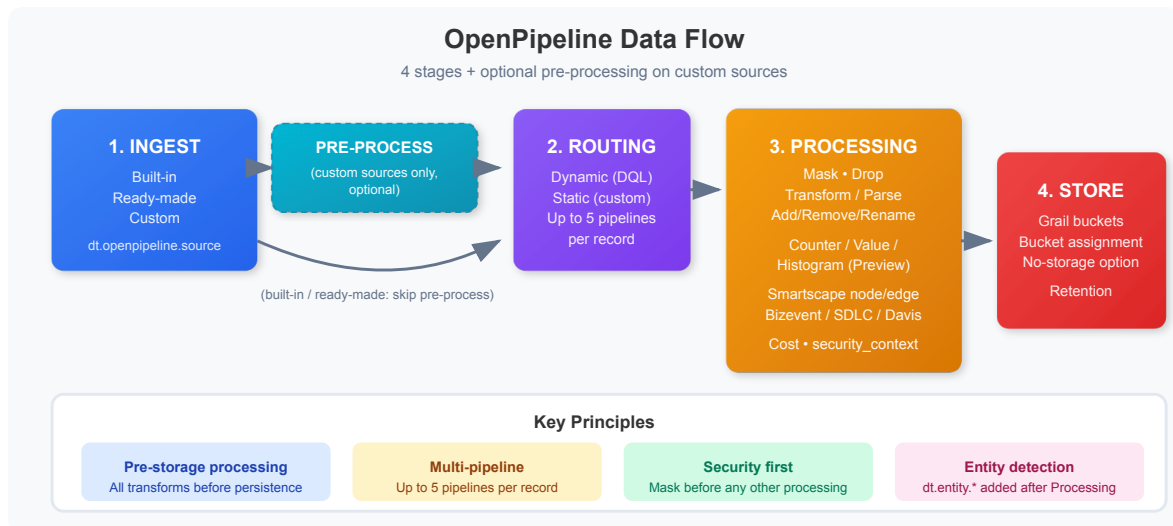
Prerequisites

Requirement	Details
Dynatrace Environment	SaaS or Managed with Grail enabled
API Access	<code>logs.read</code> token scope for DQL validation
Knowledge	OPMIG-01 or familiarity with Classic log ingestion concepts

Data Flow Architecture

Understanding how data flows through OpenPipeline is essential for designing effective pipelines.

Complete Data Flow



Doc alignment (May 2026): OpenPipeline's official `/concepts/data-flow` documentation describes a **four-stage flow** (Ingest → Routing → Processing → Storage) with optional pre-processing on custom sources. Earlier versions of this notebook described a five-stage flow with a separate "Extraction" stage; metric, event, Smartscape, and cost-allocation extraction processors actually live *within* the Processing stage.

Key Principles

- Pre-storage Processing:** All transformations happen BEFORE data is written to Grail.
- Order Matters:** Processors execute in the order defined within each pipeline.
- Multi-pipeline Support:** A single record can be processed by up to 5 pipelines.
- Entity Detection:** Entity fields (`dt.entity.*`) are added AFTER the Processing stage.
- Immutable Storage:** Once stored, data cannot be modified (mask early!).

Processing Stages in Detail

Stage 1: Ingest

Records enter via an ingest source. The source is recorded as

```
dt.openpipeline.source .
```

Source Type	Owner	Pre-processing	Routing
Built-in	OpenPipeline (view-only)	No	Dynamic only
Ready-made	Extensions	No	Dynamic only
Custom	User-defined	Yes (optional)	Dynamic or static

Stage 2: Pre-processing (*optional, custom sources only*)

A DQL transform applied to records from a custom source *before* routing. Use this to normalize raw data from third-party shippers into a common shape that all downstream pipelines can consume.

Stage 3: Routing

The routing stage determines which pipeline(s) process each incoming record.

Aspect	Description
Purpose	Match incoming data to appropriate pipelines
Timing	Before any pipeline processing
Configuration	Dynamic routing rules with matching conditions, or static assignment for custom sources
Fallback	Unmatched data goes to the default pipeline

Matching Condition Examples:

```
k8s.namespace.name == "production"
log.source == "nginx"
contains(content, "payment")
dt.openpipeline.source == "oneagent"
```

Stage 4: Processing

All in-pipeline work happens here. Processors execute in the order defined in the pipeline. Functional groups within Processing:

Group	Processors	Purpose
Masking	DQL with <code>replacePattern</code>	Redact PII before anything else
Filtering	Drop record	Remove unwanted records early
Field & record manipulation	Add/Remove/Rename fields, DQL, Parse, Technology	Transform and enrich
Metric extraction	Counter, Value, Histogram (Preview); sampling-aware variants for spans	Generate timeseries from records
Smartscape topology	Smartscape node, Smartscape edge	Populate the Smartscape topology graph
Event extraction	Business event, SDLC event, Davis event	Convert records into events
Cost & security	DPS Cost Allocation — Cost Center, DPS Cost Allocation — Product, Set <code>dt.security_context</code>	Attribute cost / apply record-level access
Storage assignment	Bucket assignment, No storage assignment	Place the record in a bucket, or skip retention

Recommended order *within* Processing: mask → drop → transform/parse → extract → assign cost/security → bucket assignment.

Stage 5: Storage

The storage stage routes data to Grail buckets. Use the **No storage assignment** processor to skip retention entirely (useful when a record is only needed to extract a metric or event).

Aspect	Description
Bucket Routing	Direct data to specific buckets
Retention	Different retention periods per bucket
Cost Control	Route high-volume data to shorter retention or to no-storage

Pipeline Types

OpenPipeline supports three types of pipelines:

1. Default Pipeline

- **Purpose:** Handles data that doesn't match any custom pipeline
- **Configuration:** Minimal processing, default bucket
- **Modification:** Can add processors but cannot delete

2. Custom Pipelines

- **Purpose:** Targeted processing for specific data sources or patterns
- **Configuration:** Full control over all processing stages
- **Routing:** Requires matching conditions in dynamic routing

3. Built-in Pipelines

- **Purpose:** Pre-configured for common technologies
- **Examples:** Kubernetes logs, AWS logs, Azure logs
- **Modification:** Can extend but not fundamentally change

 **Best Practice:** Create focused pipelines for specific use cases rather than one large pipeline with complex conditionals.

Processor Types

DQL Processor

The DQL processor uses DQL commands to transform data. Available commands:

Command	Purpose	Example
<code>fieldsAdd</code>	Add new fields	<code>fieldsAdd environment = "prod"</code>
<code>fieldsRemove</code>	Remove fields	<code>fieldsRemove sensitive_field</code>
<code>fieldsRename</code>	Rename fields	<code>fieldsRename old_name = new_name</code>
<code>parse</code>	Extract with DPL patterns	<code>parse content, "LD:prefix INT:count"</code>

fieldsAdd Examples:

```
// Static value
| fieldsAdd environment = "production"

// Conditional value
| fieldsAdd severity = if(loglevel == "ERROR", "critical", else: "normal")

// Computed value
| fieldsAdd message_length = stringLength(content)

// From existing field
| fieldsAdd short_host = substring(host.name, 0, 10)
```

Drop Processor

Removes records matching specified conditions:

```
Matching Condition: loglevel == "DEBUG"
Result: All DEBUG logs are dropped before storage
```

Technology Processors

Built-in parsers for common log formats:

Technology	Parsed Fields
Apache	client_ip, method, path, status, bytes
Nginx	Similar to Apache with nginx-specific fields
JSON	Flattens JSON structure to fields
Syslog	facility, severity, hostname, message

Dynatrace Pattern Language (DPL)

DPL is a powerful pattern matching language used in the `parse` command.

Core Matchers

Matcher	Description	Example Match
<code>INT</code>	Integer number	<code>42</code> , <code>-17</code>
<code>LONG</code>	Long integer	<code>1234567890123</code>
<code>DOUBLE</code>	Decimal number	<code>3.14</code> , <code>-0.5</code>
<code>IPADDR</code>	IPv4 or IPv6 address	<code>192.168.1.1</code> , <code>::1</code>
<code>IPV4ADDR</code>	IPv4 address only	<code>10.0.0.1</code>
<code>IPV6ADDR</code>	IPv6 address only	<code>2001:db8::1</code>
<code>TIMESTAMP</code>	Timestamp with format	<code>2024-01-15T10:30:00Z</code>
<code>LD</code>	Line data (to delimiter)	Any text until delimiter
<code>DATA</code>	Any data (greedy)	Consumes remaining text
<code>SPACE</code>	Whitespace	Spaces, tabs
<code>NSPACE</code>	Non-whitespace	Any non-space chars
<code>WORD</code>	Word characters	<code>hello</code> , <code>user123</code>
<code>JSON</code>	JSON structure	<code>{"key": "value"}</code>

Matcher	Description	Example Match
EOL	End of line	Line terminator

Pattern Syntax

Element	Syntax	Description
Export to field	<code>MATCHER:fieldname</code>	Extract and name the field
Match only	<code>MATCHER</code>	Match but don't extract
Optional	<code>MATCHER?</code>	Matcher is optional
Literal	<code>'exact text'</code>	Match literal string
Alternatives	<code>('opt1'\ 'opt2')</code>	Match either option
Quantifier	<code>MATCHER{2,5}</code>	Match 2-5 times

Quantifier caveat: `{n,m}` is not universal. `INT` accepts only `*` and `+` — `INT{3}` is rejected. For fixed-width digit runs use a character class: `[0-9]{3}`.

DPL Examples

```
// Extract IP and port from log
| parse content, "IPADDR:client_ip ':' INT:port"

// Extract user ID with flexible prefix
| parse content, "('user='| 'userId='| 'user_id=')LD:user_id"

// Parse Apache-style log
| parse content, "IPADDR:client_ip SPACE '-' SPACE LD:user SPACE '['
LD:timestamp ']"

// Extract JSON payload
| parse content, "LD JSON:payload"

// Parse with optional port
| parse content, "IPADDR:ip (':' INT:port)?"

// Extract error code
| parse content, "'error_code=' INT:error_code"
```

Timestamp Format Patterns

Symbol	Meaning	Example
yyyy	Year (4 digits)	2024
MM	Month (01-12)	01
dd	Day (01-31)	15
HH	Hour 24h (00-23)	14
mm	Minute (00-59)	30
ss	Second (00-59)	45
SSS	Milliseconds	123

Complete OpenPipeline Limits Reference

This comprehensive reference contains ALL OpenPipeline limits you need to know for planning your migration.

Data Size & Volume Limits

Limit	Value	Behavior When Exceeded
Max record size (after processing)	16 MB	Record is dropped , logged in audit
Max record size (before processing)	10 MB	Ingestion rejected with 413 error
Working memory per record	16 MB	Processing fails, record dropped
Log attribute size	32 KB	Attribute value truncated
Max field name length	255 characters	Field creation fails
Max string field length	32 KB	Content truncated
Max array size	1000 elements	Array truncated
Max nesting depth (JSON)	10 levels	Deeper levels flattened

Processing Limits

Limit	Value	Impact
Max pipelines per record	5	A record can be processed by up to 5 different pipelines. After 5, data extraction stops but the record is still persisted.
Max processors per pipeline	1,000	Cannot add more processors to pipeline
Max DQL commands per processor	10 commands	Split complex logic into multiple processors

Limit	Value	Impact
Max parse operations per processor	100 patterns	Create additional parse processors
Max fields per record	500 fields	Additional fields ignored
Processing timeout per record	30 seconds	Record dropped if exceeded
Max processor name length	100 characters	Validation error

Timestamp Constraints

Data Type	Accepted Range	Records Outside Range
Logs	24 hours past to 10 minutes future	Dropped
Spans	60 minutes past (end time)	Dropped
Events	24 hours past to 10 minutes future	Dropped
Business Events	24 hours past to 10 minutes future	Dropped
Metrics	1 hour past to 1 minute future	Dropped

 **Critical:** Historical data imports require workarounds. Contact Dynatrace support for backfilling options.

Pipeline Group Limits (*per group*)

Limit	Value	Notes
Pipeline slots per group	10	Per /reference/limits
Member pipelines per group	1,000	Per /reference/limits

Pipeline & Routing Limits

Limit	Value	Scope
Max custom pipelines	100 pipelines	Per configuration scope (logs, spans, etc.)
Max dynamic routes	3,000 routes	Per configuration scope
Max conditions per route	10 conditions	Combine with AND/OR operators
Max pipeline name length	100 characters	Validation error
Max route name length	100 characters	Validation error

Extraction Limits

Limit	Value	Notes
Max metric extractions per pipeline	10	Value + counter metrics combined
Max event extractions per pipeline	5	All event types combined
Max bizevent extractions per pipeline	3	Business events only
Max dimensions per metric	10 dimensions	Keep cardinality low
Metric key length	250 characters	Validation error
Event type name length	100 characters	Validation error

DQL & DPL Limits

Limit	Value	Context
DQL processor script length	8,192 characters	Per /reference/limits (May 2026 doc)

Limit	Value	Context
Processor matching condition length	4,096 characters (Settings API); 1,500 for legacy Configurations API / Classic pipelines	Per <code>/reference/limits</code> (Jun 17, 2026 doc)
Max DPL pattern length	4 KB	Per parse pattern
Max captured groups per parse	100 groups	Use multiple parse operations
Max alternatives in pattern	50 alternatives	<code>(opt1\ opt2\ ...\ opt50)</code>

Bucket & Storage Limits

Limit	Value	Notes
Max custom buckets	50 buckets	Per environment
Min retention period	1 day	Per bucket
Max retention period	2555 days (~7 years)	Per bucket
Bucket name length	100 characters	Alphanumeric + underscore

Rate Limits

Endpoint	Limit	Per
Log Ingest API	500 requests/min	Per token
OTLP Endpoint	1000 requests/min	Per token
Config API	100 requests/min	Per token

Field & Matching Restrictions

Read-Only Fields (Cannot be modified):

dt.ingest.*	- Ingestion metadata
dt.openpipeline.*	- Pipeline processing metadata
dt.retain.*	- Retention information
dt.system.*	- System metadata (bucket, etc.)
timestamp	- Record timestamp (can parse new, not modify original)

Reserved Field Prefixes (Cannot create):

dt.*	- Reserved for Dynatrace
dynatrace.*	- Reserved for Dynatrace

Entity Fields (Available ONLY after Processing stage):

```
dt.entity.service
dt.entity.host
dt.entity.process_group
dt.entity.process_group_instance
dt.entity.kubernetes_cluster
dt.entity.cloud_application
dt.entity.cloud_application_namespace
```

 **Design Pattern:** Entity fields are added by Dynatrace AFTER the Processing stage. You cannot:

- Use them in routing conditions
- Use them in matching conditions during processing
- Modify them with fieldsAdd/fieldsRename

You CAN use them in:

- Extraction stage (metrics, events)
- Storage stage (bucket routing based on entity)

Workarounds for Common Limit Issues

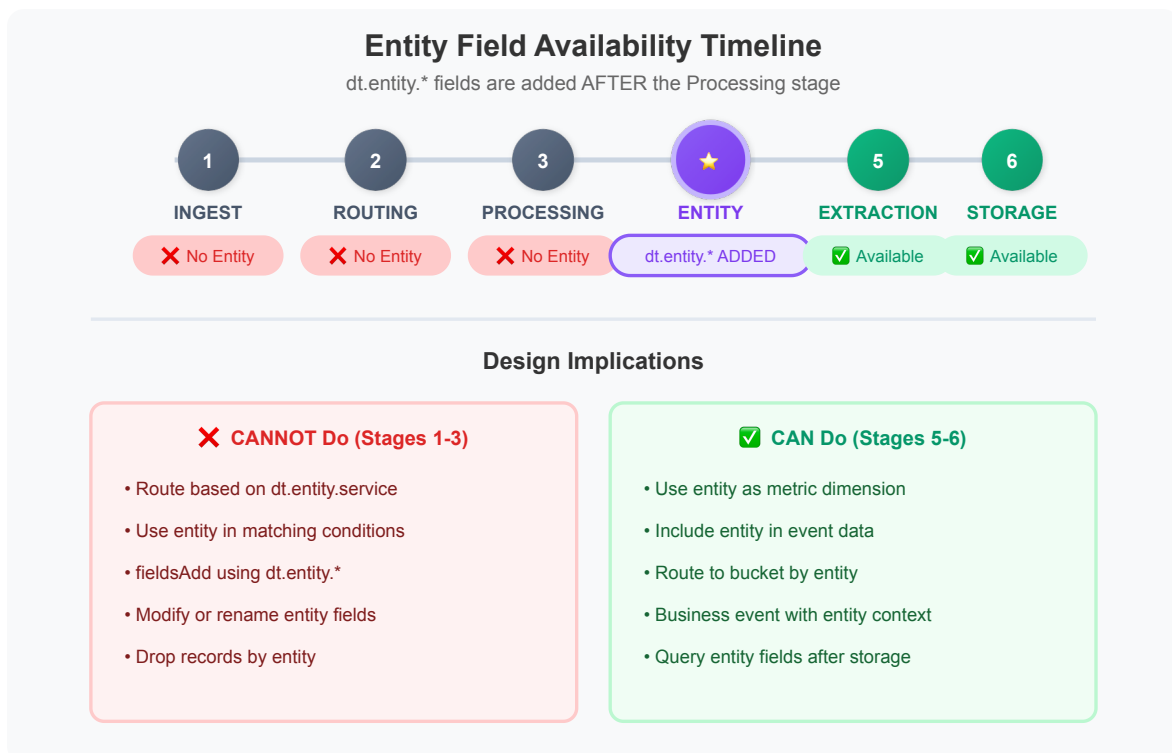
Problem	Workaround
>50 processors needed	Split into multiple pipelines, use multi-pipeline routing
>10 DQL commands	Break into multiple processors (order matters!)
>100 parse patterns	Use multiple parse processors in sequence

Problem	Workaround
>16MB after processing	Drop unnecessary fields, reduce field sizes
>5 pipelines needed	Consolidate processing logic, use conditional processors
>10 metric dimensions	Reduce cardinality, use separate metrics

Entity Field Availability Timeline

Understanding WHEN entity fields become available is critical for pipeline design.

Processing Stage Timeline



Practical Examples

❌ WRONG - Cannot route based on entity

Dynamic Route:

Condition: `dt.entity.service == "SERVICE-123"` ❌ FAILS

Why: Entity fields not available during routing

❌ WRONG - Cannot use entity in processing

DQL Processor:

fieldsAdd `service_name = dt.entity.service` ❌ FAILS

Why: Entity fields not available during processing

✅ CORRECT - Use entity in metric extraction

Metric Extraction:

Key: `log.request.duration`

Value: `duration_ms`

Dimensions: `dt.entity.service, dt.entity.host` ✅ WORKS

Why: Entity fields available in extraction stage

✅ CORRECT - Use entity for bucket routing

Bucket Routing (in Storage stage):

Condition: `dt.entity.service == "CRITICAL-SERVICE"`

Bucket: `critical_logs` ✅ WORKS

Why: Entity fields available in storage stage

Design Patterns

Pattern 1: Route by source, extract metrics by entity

Routing: `log.source == "payment-service"` (uses source field)
Processing: Parse payment data
Extraction: Create metric with `dt.entity.service` dimension ✓

Pattern 2: Enrich with custom field, then use entity

Processing: `fieldsAdd app_tier = "frontend"` (custom field)
Extraction: Dimensions: `app_tier, dt.entity.service` ✓

Pattern 3: Cannot mix entity with routing

✗ Route based on entity → Use workaround:
– Route based on source/content patterns instead
– Use service name string (not entity ID) if available

Key Fields & Metadata

OpenPipeline-Specific Fields

Field	Description	Example Value
<code>dt.openpipeline.source</code>	Data source identifier	<code>oneagent</code> , <code>generic</code> , <code>otlp</code>
<code>dt.openpipeline.pipelines</code>	Pipeline(s) that processed record	<code>["custom-pipeline-1"]</code>
<code>dt.system.bucket</code>	Grail storage bucket	<code>default_logs</code> , <code>custom_logs</code>

Log Fields

Field	Description
<code>timestamp</code>	Log timestamp (primary time field)
<code>content</code>	Log message content

Field	Description
<code>log.level</code>	Log level (ERROR, WARN, INFO, DEBUG)
<code>status</code>	Status string (alternative to loglevel)
<code>log.source</code>	Log source identifier
<code>log.iostream</code>	Stream type (stdout, stderr)

Entity Context Fields

Field	Description
<code>dt.entity.host</code>	Host entity ID
<code>dt.entity.process_group</code>	Process group entity ID
<code>dt.entity.process_group_instance</code>	Process instance entity ID
<code>dt.entity.service</code>	Service entity ID
<code>host.name</code>	Host name (string)
<code>process.executable.name</code>	Process executable name

⚠ Important: Entity fields (`dt.entity.*`) are added AFTER the Processing stage. You cannot use them in routing or processing conditions.

Exploring Your Pipeline Configuration

Use these queries to understand how your environment is configured.

```
// View data sources currently sending to OpenPipeline
// Shows the distribution of data by ingestion source
fetch logs, from: now() - 24h
| summarize {record_count = count()}, by: {dt.openpipeline.source}
| sort record_count desc
```

```
// Analyze which pipelines are processing your logs
// Helps verify routing is working correctly
fetch logs, from: now() - 24h
| filter isNotNull(dt.openpipeline.pipelines)
| summarize {record_count = count()}, by: {dt.openpipeline.pipelines}
| sort record_count desc
```

```
// Check bucket distribution for stored logs
// Verify data is routing to expected buckets
fetch logs, from: now() - 24h
| summarize {record_count = count()}, by: {dt.system.bucket}
| sort record_count desc
```

```
// Analyze pipeline processing by source and pipeline
// Shows the relationship between sources and pipelines
fetch logs, from: now() - 24h
| filter isNotNull(dt.openpipeline.pipelines)
| summarize {record_count = count()}, by: {dt.openpipeline.source,
dt.openpipeline.pipelines}
| sort record_count desc
| limit 25
```

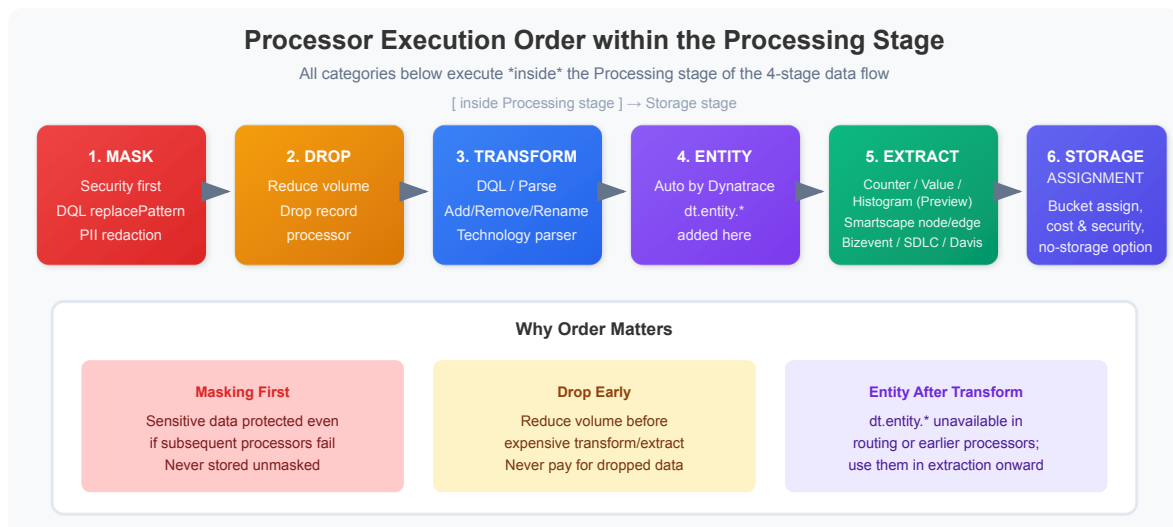
```
// Check for span processing through OpenPipeline
// Spans also flow through OpenPipeline
fetch spans, from: now() - 24h
| summarize {span_count = count()}, by: {dt.openpipeline.source}
| sort span_count desc
```

```
// View pipeline processing over time
// Helps identify volume patterns and processing trends
fetch logs, from: now() - 24h
| filter isNotNull(dt.openpipeline.pipelines)
| makeTimeseries {record_count = count()}, by: {dt.openpipeline.pipelines},
interval: 1h
```

```
// Identify logs going to default pipeline (may need custom routing)
// High volume in default may indicate missing routing rules
fetch logs, from: now() - 24h
| filter isNull(dt.openpipeline.pipelines) OR dt.openpipeline.pipelines == ""
| summarize {unrouted_count = count()}, by: {log.source}
| sort unrouted_count desc
| limit 20
```

Understanding Processing Order

The order of processors within the Processing stage is critical:



💡 **Tip:** Masking is applied FIRST so sensitive data is protected even if subsequent processors fail. The numbered groups above are processor *categories within the Processing stage* — they are not separate pipeline stages.

Summary: Key Architecture Concepts

Concept	Key Points
Data Flow	Ingest → (Pre-process) → Route → Process → Store (4 stages + optional pre-processing per /concepts/data-flow)
Pre-storage	All processing happens before data is persisted
Source Types	Built-in / Ready-made / Custom — only custom sources support pre-processing and static routing
Routing	Dynamic (DQL matcher) or static (custom sources only)
Processing Order	Mask → Drop → Transform → Extract → Cost/Security → Storage assignment
Entity Detection	Happens AFTER processing, BEFORE most extraction processors run
Multi-pipeline	One record can be processed by up to 5 pipelines (/reference/limits)
DPL	Powerful pattern language for parsing
Buckets	Control retention and cost at the storage assignment step; or skip with No storage assignment

Next Steps

Now that you understand OpenPipeline architecture, continue with:

Notebook	Focus Area
OPMIG-03	Migration Assessment & Planning
OPMIG-04	Pipeline Configuration Fundamentals
OPMIG-05	Routing & Bucket Management
OPMIG-06	Processing, Parsing & Transformation

References

- [OpenPipeline Data Flow](#)
 - [OpenPipeline Processing](#)
 - [Dynatrace Pattern Language](#)
 - [OpenPipeline Limits](#)
 - [DPL Architect Tool](#)
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